ESP32

Use boot control pins with caution: 0, 2, 5, 12, 15
IO0: Avoid device connections. Can be used a stat LED.
IO2: Avoid external pullups – will cause bootloader fail.
IO5: Has builtin pullup at POR.
IO12: Avoid external pullups – will cause bootloader fail.
IO15: Has builtin pullup at POR.

ADC2 is unavailable when WiFi is enabled.

MOMENTARY-SWITCH-5PST-SMD-5.2-TALL-REDUNDANT
10KOHM-0603-1/10W-1%
0.1UF-0603-25V-(+80/-20%)
Boot Control
ADC2
ADC1
3V3
EN
GND
TX-IO1
RX-IO3
IO0
IO2
CS3/IO5
POCI2/IO12
CS2/IO15
IO4
PICO2/IO13
SCK2/IO14
IO16
IO17
SCK3/IO18
POCI3/IO19
SDA/IO21
SCL/IO22
PICO3/IO23
DAC1/IO25
DAC2/IO26
IO27
XTAL_P/IO32
XTAL_N/IO33
IA34
IA35
SENSOR_VP/IA36
SENSOR_VN/IA39
ESP32_TXD0
ESP32_RXD0
ESP32_0
ESP32_GPIO2
ESP32_GPIO5_CS
ESP32_GPIO12_TDI
ESP32_GPIO15_TDO
ESP32_GPIO4
ESP32_GPIO13_TCK
ESP32_GPIO14_TMS
ESP32_GPIO16
ESP32_GPIO17
ESP32_GPIO18_SCK
ESP32_GPIO19_POCI
ESP32_SDA
ESP32_SCL
ESP32_GPIO23_PICO
ESP32_GPIO25
ESP32_GPIO26
ESP32_GPIO27
ESP32_ADC4
ESP32_ADC5
ESP32_ADC6
ESP32_ADC7
ESP32_ADC0
ESP32_ADC3

Fuel Gauge ALERT Connection Jumper

ESP32_GPIO4 — GAUGE_ALERT
!ALERT!O
Close jumper to enable GAUGE_ALERT to be connected to ESP32 GPIO pin 4

JTAG Header

J4 JTAG
TMS0 — ESP32_GPIO14_HDR
ESP32_GPIO14_TMS
ESP32_GPIO13_TCK
ESP32_GPIO15_TDO
ESP32_GPIO12_TDI
TCK0 — ESP32_GPIO13_HDR
Open jumpers to isolate GPIO13 and GPIO14 from the JTAG header

Auto Reset

R6 10KOHM-0603-1/10W-1%
R7 10KOHM-0603-1/10W-1%
Q1B MBT3904DW1T1
Q1A MBT3904DW1T1
DTR
RTS
ESP32_EN
ESP32_0

Boot Mode Configuration

Pin	Run	Bootloader	Default
IO0	1	0	1
IO2	x	0	0
IO5	1	x	1
IO12	0	0	0
IO15	x	x	1

IO12 = MDTI, IO15 = MDTO

If DTR is LOW, toggling RTS from HIGH to LOW resets to run mode.
If RTS is HIGH, toggling DTR from LOW to HIGH resets to bootloader.

Arduino Headers

Inputs are NOT 5V tolerant.

J5 J6 J7 J8 J9 J10 J11 J12
3.3V 5V GND
ESP32_SCL ESP32_SDA ESP32_GPIO18_SCK ESP32_GPIO19_POCI ESP32_GPIO23_PICO ESP32_GPIO5_CS ESP32_GPIO27 ESP32_GPIO26
ESP32_RXD0 ESP32_TXD0 ESP32_GPIO4 ESP32_GPIO13_HDR ESP32_GPIO14_HDR ESP32_GPIO16 ESP32_GPIO17 ESP32_GPIO25
LED_DO ESP32_EN VIN
ESP32_ADC5 ESP32_ADC4 ESP32_ADC7 ESP32_ADC6 ESP32_ADC3 ESP32_ADC0

External Power 5V Regulator

D1 3A/40V/500mV
U3 LM1117 IN OUT ADJ
VIN Range: 7-15V
C9 47UF-POLAR-TANT-SMD-7343-16V-10%
C10 100UF-POLAR-TANT-SMD-7343-16V-10%
C11 0.1UF-0603-25V-(+80/-20%)
R10 2400HM-0603-1/10W-1%
R11 715OHM-0603-1/10W-1%
5V_EXT

3.3V Regulator

U4
5V
3.3V
C12 0.1UF-0603-25V-(+80/-20%)
C13 0.1UF-0603-25V-(+80/-20%)
6V max in
700mA out

Current Measurement Header and Fuse Jumper

Open jumper to measure board current on J10
This is the penny-in-the-fuse jumper. Close only if you know what you're doing.
MEAS0 — BYP0
5V_MEAS
5V
C4 0.1UF-0603-25V-(+80/-20%)
F1 PPTC_0.75A
C18 0.1UF-0603-25V-(+80/-20%)

USB-C

USB_SOLDER_PADS
J3
J2 USB-C
VBUS D- D+ GND CC1 CC2 SHLD
SHLD0
R2 5.1K
R3 5.1K
C5 10UF-POLAR-TANT-16V-10%(TANT)
C8 0.1UF-0603-25V-(+80/-20%)
C6 10UF-0603-50V-10%
V_USB
U2 CH340C
VCC V3 R232 UD+ UD- GND TXO RXI RTS DCD RI DSR CTS DTR
R5 1KOHM-0603-1/10W-1% D3 GREEN
R4 1KOHM-0603-1/10W-1% D2 GREEN
LED_TXO
LED_RXO
Cut traces to remove TX and RX LEDs
ESP32_RXD0
ESP32_TXD0

LiPo Charger

U5 MCP73831-2-OT
VIN VBAT STAT VSS PROG
5V_CHG
C15 4.7UF-0603-35V-(20%)
R1 1KOHM-0603-1/10W-1%
D4 Yellow
R21
LED_CHG0
Cut trace to remove charge LED
R22 2KOHM-0603-1/10W-5%
C16 4.7UF-0603-35V-(20%)
V_BATT
J16 LiPo
JP2
Charge current:
1000 / R_PROG
R_PROG=2k : I_CHG=500mA

5V Bus Perfect Diodes

External power has priority over USB-C power
V_USB
R13 10KOHM-0603-1/10W-1%
R14 10KOHM-0603-1/10W-1%
Q3 Q4A Q4B Q5
R30 100KOHM-0603-1/10W-1%
R12 10KOHM-0603-1/10W-1%
Q2B Q2A
R15 10KOHM-0603-1/10W-1%
R16 47KOHM-0603-1/10W-1%
5V_EXT
5V_MEAS
Q7 Q6A Q6B Q9 Q8A Q8B
R17 100KOHM-0603-1/10W-1%
R18 1MOHM-0603-1/4W-5%
R19 100KOHM-0603-1/10W-1%
R20 1MOHM-0603-1/4W-5%

Status LED (GPIO18)

ESP32_GPIO18_SCK
D6 BLUE
Cut trace to remove status LED
R9 4.7KOHM-0603-1/10W-1%
LED_18

Boot Pushbutton

MOMENTARY-SWITCH-SPST-SMD-5.2-TALL-REDUNDANT
JUMPER-SMT_2_NO_SILK
ESP32_0
DOOTO
0.1UF-0603-25V-(+80/-20%)

RGB LED

D7 WS2812_2020
ESP32_GPIO2
DI DO

Battery Fuel Gauge

V_BATT
C17 0.1UF-0603-25V-(+80/-20%)
R23 10KOHM-0603-1/10W-1%
VDD CELL GND ALRT CTG QSTRT SDA SCL
GAUGE_ALERT
ESP32_SDA
ESP32_SCL
I2C Address: 0x36

Charger Input Perfect Diodes

V_USB
5V_EXT
5V_CHG
Q11 Q13
Q10A Q10B Q12A Q12B
R24 100KOHM-0603-1/10W-1%
R25 1MOHM-0603-1/4W-5%
R26 100KOHM-0603-1/10W-1%
R27 1MOHM-0603-1/4W-5%

Power LED

D5
R8 2.2K-0603-1/10W-1%
PWR_PWR0
Cut trace to remove power LED

Micro SD

J15
VCC DAT3/CS DAT2 DAT1 DAT0/SDO CLK/SCK CMD/SDI DET GND
ESP32_GPIO5_CS
ESP32_GPIO19_POCI
ESP32_GPIO18_SCK
ESP32_GPIO23_PICO
SD_DET0
ESP32_GPIO16
C19 10UF-0603-6.3V-20%
Close jumper to enable SD card detection

Qwiic

J13
ESP32_SCL
ESP32_SDA
SCL SDA VCC GND

I2C Pullups

I2C0
R28 2.2KOHM-0603-1/10W-1%
R29 2.2KOHM-0603-1/10W-1%
Open jumper to remove I2C pullups.

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Special Instructions

TITLE: ${PROJECTNAME}
Design by: ${DESIGNER} Based on Arduino by: M.Banzi, D.Cuartielles, T.Igoe, G.Martino, D.Mellis
Revised by: Dryw Wade
REV: ${REV}
Date: ${CURRENT_DATE}
Sheet: ${#}/${##}
KiCad Version: 7.0.6 Schematic Editor
Title:
Size: User
Date:
Rev:
KiCad E.D.A. eeschema 7.0.6
Id: 1/1